

6-mavzu. Birinchi tartibli tenglama uchun Koshi masalasi. Yechimning mavjudligi va yagonaligi haqidagi teorema (4-soat).

Raja:

1. Lipshtits sharti
2. Mavjudlik va yagonalik teoremasi

Lipshtits sharti, integral tenglama, ketma-ket yaqinlashishlar usuli

Lipshtits sharti. Aytaylik, $f(x, y)$ funksiya biror $E \subset \mathbb{R}^2$ to'plamda aniqlangan bo'lsin. Agar

$$\exists L > 0 \quad \forall \{(x, y_1), (x, y_2)\} \subset E \quad |f(x, y_1) - f(x, y_2)| \leq L |y_1 - y_2|$$

shart bajarilsa, u holda $f(x, y)$ funksiya E to'plamda y ga nisbatan (yoki y bo'yicha) **Lipshtits shartini qanoatlantiradi** deyiladi.

Jumla. Aytaylik, D ochiq yoki yopiq soha O_y o'qiga nisbatan qavariq bo'lsin, ya'ni

$$\{(x, y_1), (x, y_2)\} \subset D \Rightarrow \{(x, y_1 + \theta(y_2 - y_1)) \mid 0 < \theta < 1\} \subset D.$$

Agar $f(x, y)$ funksiya D da y bo'yicha uzluksiz va uning ichki nuqtalarida

chegaralangan $\frac{\partial f}{\partial y}$ xususiy hosilaga ega bo'lsa, u holda $f(x, y)$ funksiya D da y ga nisbatan Lipshtits shartini qanoatlantiradi.

Chekli orttirmalar haqidagi Lagranj teoremasidan foydalanib mustaqil isbotlang.

Koshi masalasi yechimining mavjudligi va yagonaligi. Ushbu

$$\begin{cases} y' = f(x, y) & (1) \end{cases}$$

$$\begin{cases} y|_{x_0} = y_0 & (2) \end{cases}$$

(K)

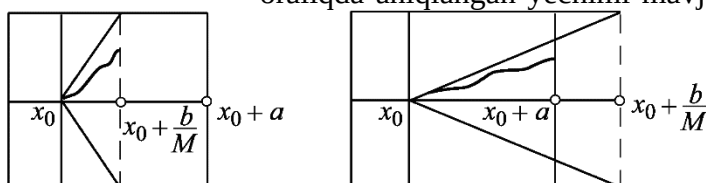
Koshi masalasiga qaytaylk. Bu masala – (x_0, y_0) nuqta orqali berilgan differensial tenglamaning integral chizig'ini o'tkazish masalasi.

Qo'yilgan (K) Koshi masalasi yechimining mavjudligi va yagonaligi uchun yetarli shart quyidagi teoremada keltirilgan.

Teorema (MYaT). Aytaylik, $f(x, y)$ funksiya ushbu $T = \{(x, y) \in \mathbb{R}^2 \mid |x - x_0| \leq a, |y - y_0| \leq b\}$ ($a > 0, b > 0$) to'rtburchakda uzluksiz va y argumentga nisbatan Lipshtits shartini qanoatlantirsin. $f(x, y)$ funksiya chegaralangan va yopiq $T \subset \mathbb{R}^2$ to'plamda uzluksiz bo'lgani uchun u shu T da chegaralangan; demak, shunday $M > 0$ son mavjudki, barcha $(x, y) \in T$ nuqtalar uchun $|f(x, y)| \leq M$ bo'ladi.

$$h = \min\{a, \frac{b}{M}\} \quad (h > 0)$$

deylik. U holda (K) Koshi masalasining $I = [x_0 - h, x_0 + h]$ oraliqda aniqlangan yechimi mavjud va bu yechim yagonadir (



1- rasm.

Isbotni bir necha qismga bo'lib bajaramiz.

1. Integral tenglamaga o'tish. Faraz qilaylik, $y = y(x)$ funksiya (K) Koshi masalasining biror $I, x_0 \in I$, oraliqda aniqlangan yechimi bo'lsin. Shu oraliqda bu funksiya (1.1) tenglamani va (1.2) shartni qanoatlantiradi:

$$\begin{cases} y'(s) = f(s, y(s)), & s \in I, \\ y(x_0) = y_0 \end{cases}$$

Bu yerdagi birinchi ayniyatning har ikkala tomoni I oraliqda uzluksiz funksiyadan iborat. Uni $s = x_0$ dan $s = x \in I$ gacha integrallaymiz va boshlang'ich shartdan foydalanamiz:

$$y(x) = y_0 + \int_{x_0}^x f(s, y(s)) ds, \quad x \in I$$

Demak, $y = y(x) \in C^1(I)$ yechim I oraliqda ushbu

$$y = y_0 + \int_{x_0}^x f(s, y(s)) ds \quad (3)$$

integral tenglamani qanoatlantiradi.

Endi faraz qilaylik, $y = y(x) \in C(I)$ funksiya (3) integral tenglamaning $I, x_0 \in I$, oraliqda yechimi bo'lsin:

$$y(x) = y_0 + \int_{x_0}^x f(s, y(s)) ds, \quad x \in I$$

Bu ayniyatning o'ng tomonidagi funksiya I oraliqda uzluksiz differensiallanuvchi (integral ostidagi funksiya uzluksiz funksiyalar kompozitsiyasi sifatida uzluksiz); demak, uning chap tomoni ham shu xususiyatga ega, ya'ni aslida $y(x) \in C^1(I)$. Bu ayniyatni differensiallab, $y(x) \in C^1(I)$ funksiya I oraliqda (1) differensial tenglamani qanoatlantirishini ko'ramiz, ayniyatda $x = x_0$ deb esa (2) boshlang'ich shartning bajarilishiga ham ishonch hosil qilamiz. Demak, (3) integral tenglamaning $y = y(x) \in C(I)$ yechimi (K) Koshi masalasining I oraliqda aniqlangan yechimidan iborat.

Shunday qilib, (K) Koshi masalasini yechish (3) integral tenglamaning $y \in C(I)$, $x_0 \in I$, yechimini topishga teng kuchlidir. Endi biz (K) masala o'rniga unga ekvivalent bo'lgan (3) integral tenglamani yechamiz.

2. Yechimga ketma-ket yaqinlashishlarni qurish. (3) integral tenglamani ketma-ket yaqinlashishlar metodi yordamida yechamiz. Boshlang'ich yaqinlashish sifatida ushbu

$$y_0(x) = y_0 \quad (4)$$

o'zgaras funksiyani tanlaymiz. Endi ketma-ket quyidagi funksiyalarni kiritamiz:

$$y_1(x) = y_0 + \int_{x_0}^x f(s, y_0(s)) ds, \quad (4_1)$$

$$y_2(x) = y_0 + \int_{x_0}^x f(s, y_1(s)) ds, \quad (4_2)$$

.....

$$y_n(x) = y_0 + \int_{x_0}^x f(s, y_{n-1}(s)) ds, \quad (4_n)$$

.....

Bu formulalardagi integrallarning mavjud bo'lishini, ya'ni $y_1(x), y_2(x), \dots, y_n(x), \dots$ ketma-ket yaqinlashishlarning aniqlangan bo'lishini ta'minlashimiz kerak.

Agar

$$|x - x_0| \leq a \quad (5)$$

bo'lsa, (4₁)dagi integral mavjud va $y_1(x)$ uzluksiz funksiyadan iborat (integral ostidagi funksiya uzluksiz). (4₂)dagi integral aniqlangan bo'lishi uchun $(s, y_1(s))$ o'zgaruvchi nuqta T to'rtburchakdan chiqib ketmasligi kerak. (4₁)ga ko'ra

$$|y_1(x) - y_0| = \left| \int_{x_0}^x f(s, y_0(s)) ds \right| \leq \left| \int_{x_0}^x |f(s, y_0)| ds \right| \leq \left| \int_{x_0}^x M ds \right| = M |x - x_0|.$$

Bundan ravshanki, $|y_1(x) - y_0| \leq b$ bo'lishi uchun $M |x - x_0| \leq b$, ya'ni

$$|x - x_0| \leq \frac{b}{M} \quad (6)$$

shartning bajarilishi yetarli. Demak, (5) va (6) shartlar birgalikda o'rinli, ya'ni

$|x - x_0| \leq h$ ($h \leq a$ va $h \leq \frac{b}{M}$) bo'lganda $(x, y_1(x)) \in T$ bo'ladi. Bundan keyin x o'zgaruvchi uchun ana shu $|x - x_0| \leq h$ shart bajarilgan deb hisoblaymiz. Endi tushunarliki, agar $y_k(x)$ funksiya $|x - x_0| \leq h$ oraliqda aniqlangan, grafigi T da joylashgan va uzluksiz bo'lsa, $y_{k+1}(x)$ funksiya ham shu xususiyatlarga ega bo'ladi, chunki

$$\begin{aligned}
|y_{k+1}(x) - y_0| &= \left| \int_{x_0}^x f(s, y_k(s)) ds \right| \leq \\
&\leq \left| \int_{x_0}^x f(s, y_k) | ds \right| \leq \left| \int_{x_0}^x M ds \right| = M |x - x_0| \leq Mh \leq b
\end{aligned}$$

Demak, matematik induksiya prinsipiga ko'ra $y_n(x)$ ($n \in \mathbf{N}$) funksiyalarning barchasi $[x_0 - h, x_0 + h]$ oraliqda aniqlangan, grafiklari T da joylashgan va uzluksiz (aslida $y'_n(x)$ hosilalar ham uzluksiz, chunki (4_n) formulada integral ostidagi funksiya uzluksiz).

3. Ketma-ket yaqinlashishlarning tekis yaqinlashuvchiligi. Yuqorida aniqlangan $y_0(x), y_1(x), y_2(x), \dots, y_n(x), \dots$ uzluksiz funksiyalardan tuzilgan ketma-ketlikning $[x_0 - h, x_0 + h]$ oraliqda tekis yaqinlashuvchi ekanligini ko'rsatamiz. Buning uchun ushbu

$$y_0(x) + (y_1(x) - y_0(x)) + (y_2(x) - y_1(x)) + \dots \quad (7)$$

funksional qatorning tekis yaqinlashishini asoslash kerak. Tekis yaqinlashish haqidagi

Veyershtass alomatidan foydalanamiz: agar $|u_n(x)| \leq c_n, x \in I, n \in \mathbf{N}$ va $\sum_n c_n$ sonli qator yaqinlashuvchi bo'lsa, u holda $\sum_n u_n(x)$ qator I oraliqda tekis yaqinlashuvchi bo'ladi.

Teoremaning shartiga ko'ra $f(x, y)$ funksiya y o'zgaruvchiga nisbatan Lipshtits shartini qanoatlantiradi:

$$\exists L > 0 |f(x, u) - f(x, v)| \leq L |u - v| ((x, u) \in T, (x, v) \in T). \quad (8)$$

Quyidagi baholashlarni bajaramiz.

$$\begin{aligned}
|y_0(x)| &= |y_0|, \\
|y_1(x) - y_0(x)| &\leq \left| \int_{x_0}^x f(s, y_0) | ds \right| \leq M |x - x_0| \\
&\quad (\text{chunki } T \text{ da } |f(x, y)| \leq M), \quad (9_1)
\end{aligned}$$

$$\begin{aligned}
|y_2(x) - y_1(x)| &= \left| \int_{x_0}^x (f(s, y_1(s)) - f(s, y_0(s))) ds \right| \leq \\
&\leq \left| \int_{x_0}^x f(s, y_1(s)) - f(s, y_0(s)) | ds \right| \stackrel{(8)}{\leq} \left| \int_{x_0}^x L |y_1(s) - y_0(s)| ds \right| \stackrel{(9_1)}{\leq} \\
&\stackrel{(9_1)}{\leq} \left| \int_{x_0}^x LM |s - x_0| ds \right| = ML \frac{|x - x_0|^2}{2}, \quad (9_2)
\end{aligned}$$

$$|y_3(x) - y_2(x)| \leq \left| \int_{x_0}^x f(s, y_2(s)) - f(s, y_1(s)) | ds \right| \stackrel{(8)}{\leq}$$

$$\begin{aligned} & \stackrel{(8)}{\leq} \left| \int_{x_0}^x L |y_2(s) - y_1(s)| ds \right| \stackrel{(9_2)}{\leq} \\ & \stackrel{(9_2)}{\leq} \left| \int_{x_0}^x LML \frac{|s - x_0|^2}{2} ds \right| = ML^2 \frac{|x - x_0|^3}{3!}, \end{aligned} \quad (9_3)$$

$$|y_n(x) - y_{n-1}(x)| \leq ML^{n-1} \frac{|x - x_0|^n}{n!}, \quad (9_n)$$

Demak,

$$|x - x_0| \leq h \text{ bo'lganda } |y_n(x) - y_{n-1}(x)| \leq ML^{n-1} \frac{h^n}{n!}$$

va ushbu $\sum_n ML^{n-1} \frac{h^n}{n!}$ sonli qator yaqinlashuvchi bo'lgani uchun (Dalamber alomatiga ko'ra $D = 0 < 1$) (1.7) funksional qator I oraliqda tekis yaqinlashuvchi.

4. Ketma-ket yaqinlashishlar limitining yechim ekanligi. Biz uzluksiz funksyalardan tuzilgan $y_0(x), y_1(x), y_2(x), \dots, y_n(x), \dots$ ketma-ketlikning I oraliqda tekis yaqinlashuvchi ekanligini ko'rsatdik. Demak, analizdan ma'lum teorema ko'ra (uzluksiz funksyalarining tekis limiti uzluksiz)

$$\phi(x) = \lim_{n \rightarrow \infty} y_n(x), \quad |x - x_0| \leq h, \quad (10)$$

limit funksiya I oraliqda uzluksiz. Uning (3) integral tenglamani qanoatlantirishini ko'rsatamiz. (4_n) formulaga qaytaylik:

$$y_n(x) = y_0 + \int_{x_0}^x f(s, y_{n-1}(s)) ds$$

Agar

$$n \rightarrow \infty \text{ bo'lganda } \int_{x_0}^x f(s, y_{n-1}(s)) ds \rightarrow \int_{x_0}^x f(s, \phi(s)) ds \quad (11)$$

ekanligini ko'rsatsak, u holda (4_n) tenglikda limitga o'tib, (10) funksiyaning (3) integral tenglama yechimi ekanligini asoslagan bo'lamiz:

$$\phi(x) = y_0 + \int_{x_0}^x f(s, \phi(s)) ds, \quad x \in I. \quad (12)$$

(11) dagi tasdiq (8) Lipshits sharti va (10) dagi yaqinlashishning tekis ekanligidan kelib chiqadi: yetarlicha katta n nomerlar uchun barcha $x \in I$ nuqtalarda $|y_{n-1}(x) - \phi(x)| < \varepsilon$ bo'lganligiga ko'ra

$$\left| \int_{x_0}^x f(s, y_{n-1}(s)) ds - \int_{x_0}^x f(s, \varphi(s)) ds \right| \leq \left| \int_{x_0}^x |f(s, y_{n-1}(s)) - f(s, \varphi(s))| ds \right| \leq$$

$$\leq \left| \int_{x_0}^x L |y_{n-1}(s) - \varphi(s)| ds \right| < L\varepsilon |x - x_0| \leq Lh\varepsilon.$$

5. Yechimning yagonaligi. Biz (1.3) integral tenglamaning $y = \phi(x)$, $x \in I$, yechimini topdik. Uning har qanday yechimi ana shu yechim bilan ustma-ust tushishini ko'rsatamiz.

(3) tenglamaning ixtiyoriy $y = \psi(x)$, $x \in I$, yechimini qaraylik. Demak,

$$\psi(x) \in C(I) \text{ va } \psi(x) = y_0 + \int_{x_0}^x f(s, \psi(s)) ds, \quad x \in I. \quad (13)$$

$y = \phi(x)$ yechimga intiluvchi $y_n(x)$ ketma-ket yaqinlashishlar bilan $y = \psi(x)$ yechim orasidagi farqning modulini baholaymiz. Quyidagilarga egamiz:

$$|y_0(x) - \psi(x)| \stackrel{(4_0), (13)}{=} \left| \int_{x_0}^x f(s, \psi(s)) ds \right| \leq$$

$$\leq \left| \int_{x_0}^x |f(s, \psi(s))| ds \right| \leq M |x - x_0|, \quad (14_1)$$

$$|y_1(x) - \psi(x)| \stackrel{(4_1), (13)}{=} \left| \int_{x_0}^x (f(s, y_0(s)) - f(s, \psi(s))) ds \right| \leq$$

$$\leq \left| \int_{x_0}^x |f(s, y_0(s)) - f(s, \psi(s))| ds \right| \stackrel{(8) (8)}{\leq} \left| \int_{x_0}^x L |y_0(s) - \psi(s)| ds \right| \stackrel{(14_1)}{\leq}$$

$$\stackrel{(14_1)}{\leq} \left| \int_{x_0}^x LM |s - x_0| ds \right| = ML \frac{|x - x_0|^2}{2}, \quad (14_2)$$

.....

$$|y_n(x) - \psi(x)| \leq ML^n \frac{|x - x_0|^{n+1}}{(n+1)!}, \quad x \in I. \quad (14_n)$$

Oxirgi tengsizlikda $x \in I$ nuqtani tayinlab, $n \rightarrow \infty$ da limitga o'tamiz. U holda

$$\frac{a^n}{n!} \rightarrow 0$$

bo'lganligi uchun quyidagilarga ega bo'lamiz:

$$|\varphi(x) - \psi(x)| \leq 0, \quad x \in I \Rightarrow \varphi(x) \equiv \psi(x), \quad x \in I. \quad \square$$

Eslatma. (14_n) tengsizlikdan (K) Koshi masalasining yechimi $y = \phi(x)$ va unga ketma-ket yaqinlashishlar orasidagi farqning baholanishi kelib chiqadi:

$$|y_n(x) - \phi(x)| \leq ML^n \frac{|x - x_0|^{n+1}}{(n+1)!}, \quad x \in I.$$

Misol. Noma'lum $y = y(x)$ funksiyaga nisbatan qo'yolgan ushbu

$$\begin{cases} y' = y \\ y(0) = 1 \end{cases}$$

Koshi masalasini ketma-ket yaqinlashishlar metodi yordamida yeching.

Berilgan tenglamaning o'ng tomonidagi $f(x, y) = y$ funksiya MYaT shartlarini ixtiyoriy $T = \{(x, y) \in \mathbb{R}^2 \mid |x| \leq a, |y - 1| \leq b\}$ ($a > 0, b > 0$) to'rtburchakda qanoatlantiradi. Ravshanki, qaralayotgan misol uchun $M = \sup_T |f(x, y)| = \sup_T |y| = 1 + b$ va $h = \min \left\{ a, \frac{b}{1+b} \right\} < 1$. Berilgan Koshi masalasi ushbu

$$y = 1 + \int_0^x y(s) ds$$

integral tenglamaga ekvivalent. Bu tenglamani ketma-ket yaqinlashishlar metodi yordamida yechamiz:

$$\begin{aligned} y_0(x) &= 1, \\ y_1(x) &= 1 + \int_0^x y_0(s) ds = 1 + \int_0^x 1 ds = 1 + x, \\ y_2(x) &= 1 + \int_0^x y_1(s) ds = 1 + \int_0^x (1 + s) ds = 1 + x + \frac{x^2}{2!}, \\ y_3(x) &= 1 + \int_0^x y_2(s) ds = 1 + \int_0^x \left(1 + s + \frac{s^2}{2!}\right) ds = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}, \\ &\dots\dots\dots \\ y_n(x) &= 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!}, \\ &\dots\dots\dots \end{aligned}$$

Teoremada isbotlandiki, bu $y_n(x)$ ketma-ket yaqinlashishlar berilgan masalaning

yagona yechimi bo'lmish $y = e^x$ funksiyaga $|x| \leq h$, $h = \min \left\{ a, \frac{b}{1+b} \right\} < 1$, oraliqda tekis yaqinlashadi:

$$e^x = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \dots$$

Oxirgi formula, analizdan ma'lumki, aslida ixtiyoriy $x \in \mathbb{R}$ uchun o'rinli (bu yerdagi qator ixtiyoriy chegaralangan oraliqda tekis yaqinlashuvchi bo'ladi). $\square$

Eslataylikki, agar $D \subset \mathbb{R}^2$ sohaning har bir nuqtasidan $y' = f(x, y)$ differensial tenglamaning yagona integral chizig'i (yechimi) o'tsa, D soha shu tenglama uchun yagonalik sohasi deyiladi.

Quyidagi teorema isbotlangan teoremaning bevosita natijasidir.

Teorema (soha uchun MYaT). Agar $f(x, y)$ funksiya $D \subset \mathbb{R}^2$ sohada uzluksiz va ixtiyoriy $(x_0, y_0) \in D$ nuqtaning yetarlicha kichik atrofida y bo'yicha

Lipshits shartini qanoatlantirsa, u holda D soha $y' = f(x, y)$ differensial tenglamaning yagonalik sohasi bo'ladi. Xususan, agar D sohada $f(x, y)$ funksiya va uning $f'_y(x, y)$ hosilasi uzluksiz bo'lsa, D soha $y' = f(x, y)$ differensial tenglama uchun yagonalik sohasidir.

Misol. Ushbu $y' = x^2 y^3 - \ln y + 1$ differensial tenglama uchun biror yagonalik sohasini ko'rsating.

Ravshanki, $f(x, y) = x^2 y^3 - \ln y + 1$ funksiya va uning $f'_y(x, y) = 3x^2 y^2 - \frac{1}{y}$ hosilasi $y > 0$ yarim tekislikda uzluksiz. Demak, shu $y > 0$ yarim tekislik berilgan tenglama uchun yagonalik sohasidir, ya'ni $y > 0$ yarim tekislikning har bir nuqtasidan $y' = x^2 y^3 - \ln y + 1$ differensial tenglamaning yagona yechimi o'tadi.

Agar $y' = f(x, y)$ tenglamaning o'ng tomonidan faqatgina uzluksiz bo'lishni talab qilsak, u holda berilgan nuqtadan o'tuvchi yechim mavjud bo'ladi, lekin umumiy holda yechimning yagonalik xossasi bo'lmaydi.

Teorema (Peano). Agar $f(x, y)$ funksiya D sohada uzluksiz bo'lsa, D sohaning har bir nuqtasidan $y' = f(x, y)$ tenglamaning kamida bitta yechimi o'tadi.

Bu teoremani isbotlamaymiz. Isboti analizning nozik tushunchalariga tayanadi.

Agar $f(x, y)$ funksiya uzluksizlikdan boshqa shart talab qilmasak, nuqtadan o'tuvchi yechimning yagonaligi haqida hech narsa deb bo'lmaydi.

M. A. Lavrentev tekislikda uzluksiz bo'lgan shunday $f(x, y)$ funksiya qurganki, u orqali tuzilgan $y' = f(x, y)$ differensial tenglamaning $\forall (x_0, y_0) \in \mathbb{R}^2$ nuqta orqali cheksiz ko'p yechimi o'tadi.

Mavzuni mustahkamlash uchun savollar:

1. Lipshits sharti deb nimaga aytiladi?
2. Mavjudlik va yagonalik teoremasini isbotlang?
3. Birinchi tartibli tenglamaga Koshi masalasini qo'ying?